C – Dynamic Memory Allocation

Allocating Memory

- malloc(size) -> request a certain number of bytes of memory
  - on the heap
  - contiguous
  - returns the address of start of block
  - needs to be cast to appropriate pointer
- memory for one int:

```
int *p;
p = (int *) malloc(sizeof(int));
```

Allocating More Memory

- Use malloc, but request more bytes
- Often used to malloc memory for an array

```
int *p;
p = (int *) malloc(sixeof(int)*100);
```

Allocating More Memory

- calloc(count, size)
  - count -> number of items
  - size -> size of item
  - allocated contiguously
  - initialized to 0

```
int p*;
p = (int *) calloc(100, sizeof(int));
```

Allocating More Memory

- realloc(ptr, sizw)
  - ptr -> current pointer to memory trying to resize
  - size -> overall size wanted
- may not be enough room to enlarge
  - creates new, copies to new, frees the old
  - additional not guaranteed to be 0 filled

```
int *p = (int *) malloc(sizeof(int)*10);
p = (int *) realloc(ptr, sizeof(int)*20);
```

Freeing Memory

- free(ptr)

```
int p* = (int *) malloc(sixeof(int)*10);
free(p);
```